FROM RUNTIME TO REQUEST: MAPPING REAL-WORLD INPUTS TO CEDAR

Cedar expects each authorization request to follow the PARC model—Principal, Action, Resource, and Context—usually represented in JSON. However, in real-world applications, these inputs don't always start out as JSON. They often originate from sources such as the following:

- **gRPC calls**, where caller identity and action are embedded in metadata.
- **HTTP headers**, such as bearer tokens, session information, or request paths.
- **Application context**, such as device fingerprinting, current time, or environment variables.

In practice, these details must be extracted and normalized before they can be evaluated. For example, a service might:

- Pull the authenticated principal from a JWT in an HTTP request header
- Determine the action based on the endpoint being called
- Map the resource from the request body or URL path
- Derive context from headers (for example, User-Agent) or internal system state

The system invoking Cedar is responsible for constructing this request. Cedar doesn't specify how the data should be collected—only that, once it is, it must be organized according to the PARC model. This flexibility allows Cedar to easily integrate with many different architectures and protocols, without requiring a specific transport or format.

The JSON illustrates how ACME could model an access request for its Customer Collaboration platform. The policy shown in figure 7.1 depends on a few details about how ACME models its data.